Greenhouse Control

Scenario B, Round 1 - 60 minutes, teams of 4 to 5

Your task

Your team builds a temperature control for a small greenhouse. A sensor measures the air temperature. When it gets warm, a fan switches on. When it gets too hot, a warning light switches on.

In this round no standard is given. Your team decides which pins to use and which temperatures count as warm and hot.

What the system must do

- The temperature is read all the time and shown in the Serial Monitor.
- The fan LED switches on above the warm temperature you choose.
- The warning LED switches on above the hot temperature you choose.
- Bonus. The fan switches off again only a little below the warm temperature, so it does not flicker.

Parts

Part	Quantity
Arduino Uno and breadboard	1
TMP36 temperature sensor	1
LED	2
220 Ohm resistor	2
Jumper wires	about 10

Wiring

Part	Connection
TMP36 left leg	5V
TMP36 middle leg	A0
TMP36 right leg	GND
Fan LED	A digital pin your team chooses, through a 220 Ohm resistor to GND
Warning LED	A digital pin your team chooses, through a 220 Ohm resistor to GND

The code

You can build the circuit on a real board or in TinkerCad Circuits. In TinkerCad, open the code editor, switch it to Text and paste the code. Fill in every blank marked ___ before you start the program. The code you get is not fully correct. When the system does not do what the list above asks, find the cause and fix it.

Blank	What to fill in
PIN_LED_FAN and PIN_LED_WARN	The pins your team chose for the two LEDs.
FAN_ON_C	The temperature in degrees Celsius above which the fan switches on.
FAN_OFF_C	Bonus. The temperature below which the fan switches off again. If you skip the bonus, use the same value as FAN_ON_C.
WARN_C	The temperature above which the warning switches on.

```
/*
 * Scenario B  Greenhouse control
 * Round 1  Temperature control
 * Your team decides every ___ before you upload.
 */

const int   PIN_TEMP     = A0;  // TMP36, middle leg
const int   PIN_LED_FAN  = ___; // LED stands in for the fan
const int   PIN_LED_WARN = ___;

const float FAN_ON_C  = ___;   // fan turns on above this temperature
const float FAN_OFF_C = ___;   // bonus, fan turns off only below this, else same as FAN_ON_C
const float WARN_C    = ___;   // warning turns on above this temperature

bool fanOn = false;

float readTempC() {
  float v = analogRead(PIN_TEMP) * (5.0 / 1023.0);
  return (v - 0.5) * 100.0;
}

void setup() {
  pinMode(PIN_LED_FAN, OUTPUT);
  pinMode(PIN_LED_WARN, OUTPUT);
  Serial.begin(9600);
}

void loop() {
  float t = readTempC();
  if (t > FAN_ON_C)  fanOn = true;
  if (t < FAN_OFF_C) fanOn = false;
  digitalWrite(PIN_LED_FAN,  fanOn ? HIGH : LOW);
  digitalWrite(PIN_LED_WARN, t > FAN_ON_C ? HIGH : LOW);
  Serial.print("Temp "); Serial.println(t);
  delay(500);
}
```

How you work

Every member of the team takes one role.

Role	Responsibility	Works closely with
Coordinator	Keeps track of time and signs off the result.	Everyone
Sensor developer	Reads the temperature and converts it to degrees Celsius.	Actuator and display developers
Actuator developer	Switches the fan.	Sensor developer

Role	Responsibility	Works closely with
Display and report developer	Switches the warning light and prints the temperature.	Sensor developer
Tester	Tests the system against the list above and reports what does not work.	Everyone

In a team of four, the coordinator also takes the role of tester.

Apart from the roles there are no rules for this round. Agree everything else as you go.

You have 60 minutes.

After the round

Take about 10 minutes as a team and talk about these questions.

- How did you decide on the pins and the temperatures?
- How often did you have to agree on something again because someone had changed it?
- Who ended up owning which part of the code?
- When something did not work, how did you find the cause?
- What would you change if you did this again?